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MATH1061

work submission coversheet

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Submission Topic Assignment 1

Due date of submission 27/08/2020

Due time of submission 1:00 PM

Group

Actual submission date _____

Name of tutor (if appropriate) _____

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Signed ....Xianglong LIN.... . Date ....26/08/2020....

Assignment 1

Q1.

p	q	r	$p \to q$	$\sim r \to \sim q$	$\sim(r \wedge p)$	$\sim(p \vee q)$	$\sim(p \wedge r)$
T	T	T	T	T	F	F	F
T	T	F	T	F	T	F	F
T	F	T	F	T	T	F	F
F	T	T	T	T	T	F	T
T	F	F	F	T	T	F	F
F	F	T	T	T	T	T	F
F	T	F	T	F	T	F	F
F	F	F	T	T	T	T	F

$(p \to q) \wedge (\sim r \to \sim q) \wedge \sim(\sim p)$	$(\sim(p \vee q) \vee (\sim p \wedge r))$	$(\sim(p \to q) \wedge (\sim r \to \sim q) \wedge \sim(r \wedge p)) \leftrightarrow (\sim(p \vee q) \vee (\sim p \wedge r))$
F	F	F
F	F	F
F	F	F
T	T	F
F	F	F
T	T	F
F	F	F
T	T	F

∴ The statement is a contradiction

$((p \to q) \wedge (\sim r \to \sim q) \wedge \sim(r \wedge p)) \leftrightarrow (\sim(p \vee q) \vee (\sim p \wedge r))$

$\equiv ((\sim p \vee q) \wedge (r \vee \sim q) \wedge \sim(r \wedge p)) \leftrightarrow (\sim(p \vee q) \vee (\sim p \wedge r))$

(Since $p \to q \equiv \sim p \vee q$)

$\equiv ((\sim p \vee q) \wedge (r \vee \sim q) \wedge (\sim r \vee \sim p)) \leftrightarrow ((\sim p \wedge \sim q) \vee (\sim p \wedge r))$

(De Morgan's Laws)

$\equiv ((\sim p \vee q) \wedge (\sim r \vee \sim p) \wedge (r \vee \sim q)) \leftrightarrow ((\sim p \wedge \sim q) \vee (\sim p \wedge r))$

(Associative Laws)

$\equiv (\sim p \vee (q \wedge \sim r) \wedge (r \vee \sim q)) \leftrightarrow (\sim p \wedge (\sim q \vee r))$

(Distributive Laws)

$\equiv (\sim p \vee (q \wedge \sim r) \wedge (q \wedge \sim r)) \leftrightarrow (\sim p \wedge (\sim q \vee (q \wedge \sim r)))$

(De Morgan's Laws)

$\equiv (q \wedge \sim r) \leftrightarrow (\sim p \wedge (\sim q \vee (q \wedge \sim r)))$

(Absorption laws)

$\equiv (q \wedge \sim r) \leftrightarrow \sim(p \vee (q \wedge \sim r))$

(De Morgan's Laws)

$\equiv ((q \wedge \sim r) \to \sim(p \vee (q \wedge \sim r))) \wedge (\sim(p \vee (q \wedge \sim r)) \to (q \wedge \sim r))$

(Since $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$)

$\equiv ((\sim(q \wedge \sim r)) \vee (\sim(p \vee (q \wedge \sim r)))) \wedge ((p \vee (q \wedge \sim r)) \vee (q \wedge \sim r))$

(Since $p \to q \equiv \sim p \vee q$)

$\equiv ((\sim(q \wedge \sim r)) \vee (\sim(p \vee (q \wedge \sim r)))) \wedge ((p \vee (q \wedge \sim r)) \vee (q \wedge \sim r))$

(Idempotent Laws)

$\equiv ((\sim(q \wedge \sim r)) \vee (\sim p \wedge (\sim(q \wedge \sim r)))) \wedge ((p \vee (q \wedge \sim r)) \vee (q \wedge \sim r))$

(Distribution Laws)

$\equiv \sim p \wedge p$

(Absorption Laws)

$\equiv$

(Negation Laws)

$\therefore$ The statement is a contradiction

Q2.

1. $p \to q$
2. $\sim(q \vee r)$
3. $s \to r$
4. $\sim q \wedge \sim r$ (from 2 by De Morgan's Laws)
5. $\sim r$ (from 4 by Specialization)
6. $\sim s$ (from 3, 4 by Modus Tollens)

Q3

p	q	$p \nrightarrow q$	$p \Rightarrow q$	$\sim(p \Rightarrow q)$
T	T	F	T	F
T	F	T	F	T
F	T	F	T	F
F	F	F	T	F

$\therefore p \nrightarrow q \equiv \sim(p \Rightarrow q)$

(3) $p \not\to q \equiv \sim(p \to q)$ (Since $p \not\to q \equiv \sim(p \to q)$)
$\equiv \sim(\sim p \vee q)$ (Since $p \to q \equiv \sim p \vee q$)
$\equiv p \wedge \sim q$ (De Mogan's Laws)

(a) $\sim p \not\to q \equiv \sim(\sim p \to q)$ (Since $p \not\to q \equiv \sim(p \to q)$)
$\equiv \sim(\sim\sim p \vee q)$ (Since $p \to q \equiv \sim p \vee q$)
$\equiv \sim p \wedge \sim q$ (De Mogan's Laws)

(a) $p \not\to \sim q \equiv \sim(p \to \sim q)$ (Since $p \not\to q \equiv \sim(p \to q)$)
$\equiv \sim(\sim p \vee \sim\sim q)$ (Since $p \to q \equiv \sim p \vee q$)
$\equiv p \wedge q$ (De Mogan's Laws)

$\sim p \not\to \sim q \equiv \sim(\sim p \to \sim q)$ (Since $p \not\to q \equiv \sim(p \to q)$)
$\equiv \sim(p \vee \sim q)$ (Since $p \to q \equiv \sim p \vee q$)
$\equiv \sim p \wedge q$ (De Mogan's Laws)

(a) $p \vee q \equiv \sim(\sim p \wedge \sim q)$ (De Morgan's Laws)
$\equiv \sim(\sim(\sim p \vee q))$ (De Morgan's Laws)
$\equiv \sim(\sim(\sim p \to q))$ (Since $p \to q \equiv \sim p \vee q$)
(b) $\equiv \sim(\sim p \not\to q)$ (Since $p \not\to q \equiv \sim(p \to q)$)

(b) $p \wedge q \equiv \sim(\sim p \vee \sim q)$ (De Mogan's Laws)
$\equiv \sim(p \to \sim q)$ (Since $p \to q \equiv \sim p \vee q$)
$\equiv p \not\to \sim q$ (Since $p \not\to q \equiv \sim(p \to q)$)

$((p \to q) \wedge (\sim r \to \sim q) \wedge \sim(r \wedge p)) \leftrightarrow (\sim(p \vee q) \vee (\sim p \wedge r))$

$\equiv ((\sim p \vee q) \wedge (r \vee \sim q) \wedge \sim(r \wedge p)) \leftrightarrow (\sim(p \vee q) \vee (\sim p \wedge r))$

(Since $p \to q \equiv \sim p \vee q$)

$\equiv ((\sim p \vee q) \wedge (r \vee \sim q) \wedge (\sim r \vee \sim p)) \leftrightarrow ((\sim p \wedge \sim q) \vee (\sim p \wedge r))$

(De Morgan's Laws)

$\equiv ((\sim p \vee q) \wedge (\sim r \vee \sim p) \wedge (r \vee \sim q)) \leftrightarrow ((\sim p \wedge \sim q) \vee (\sim p \wedge r))$

(Associative Laws)

$\equiv (\sim p \vee (q \wedge \sim r)) \wedge (r \vee \sim q)) \leftrightarrow (\sim p \wedge (\sim q \vee r))$

(Distributive Laws)

$\equiv ((\sim p \vee (q \wedge \sim r)) \wedge (q \wedge \sim r)) \leftrightarrow (\sim p \wedge (\sim q \wedge \sim r)))$

(De Morgan's Laws)

$\equiv (q \wedge \sim r) \leftrightarrow (\sim p \wedge (\sim q \wedge \sim r))$

(Absorption laws)

$\equiv (q \wedge \sim r) \leftrightarrow \sim(p \vee (q \wedge \sim r))$

(De Morgan's Laws)

$\equiv ((q \wedge \sim r) \to \sim(p \vee (q \wedge \sim r))) \wedge (\sim(p \vee (q \wedge \sim r)) \to (q \wedge \sim r))$

(Since $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$)

$\equiv ((\sim(q \wedge \sim r)) \vee (\sim(p \vee (q \wedge \sim r)))) \wedge ((p \vee (q \wedge \sim r)) \vee (q \wedge \sim r))$

(Since $p \to q \equiv \sim p \vee q$)

$\equiv ((\sim(q \wedge \sim r)) \vee (\sim(p \vee (q \wedge \sim r)))) \wedge ((p \vee (q \wedge \sim r)) \vee (q \wedge \sim r))$

(Idempotent Laws)

$\equiv ((\sim(q \wedge \sim r)) \vee (\sim p \wedge (\sim(q \wedge \sim r)))) \wedge ((p \vee (q \wedge \sim r)) \vee (q \wedge \sim r))$

(Distribution Laws)

$\equiv \sim p \wedge p$

(Absorption Laws)

$\equiv$

(Negation Laws)

$\therefore$ The statement is a contradiction

Q2. 1. $p \to q$

2. $\sim(q \vee r)$

3. $s \to r$

4. $\sim q \wedge \sim r$ (from 2 by De Morgan's Laws)

5. $\sim r$ (from 4 by Specialization)

6. $\sim s$ (from 3, 4 by Modus Tollens)

Q5. (a) False. Take $x=1 \in \mathbb{Z}$.

Then $x^2=x=1$ $\therefore p(x)$

But $x>0$ $\therefore \sim Q(x)$

Thus the statement is false.

(b) False. Take $x=0 \in \mathbb{Z}$.

Then $x^2=x=0$ $\therefore p(x)$

But $x \leq 0$ $\therefore Q(x)$

Thus the statement is false.

(c) False. $\because R(x)$ $\therefore x^2=x+1$

$\therefore x^2-x-1=0$

$a=1$ $b=-1$ $c=-1$ $\Delta=b^2-4ac=1+4=5>0$

$x=\frac{-b\pm\sqrt{\Delta}}{2a}$

$=\frac{1\pm\sqrt{5}}{2} \notin \mathbb{Z}$

Thus the statement is false.

(d) False. Take $x=1$

Then $x^2=x=1$ $\therefore p(x)$

But $x^2=1 \neq x+1=2$ $\therefore \sim R(x)$

Thus the statement is false.

(e) True.

Proof: $\because P(x) \wedge S(x)$

$\therefore x^2=x$

$x^2-x=0$

$x(x-1)=0$

$x_1=0$ $x_2=1$

Also $\because x$ is even

$\therefore x=0 \leq 0$

$\therefore Q(x)$ $\therefore \forall x \in \mathbb{Z}, (P(x) \wedge S(x)) \rightarrow Q(x)$

Thus the statement is true. $\square$

(f) True

Proof: $\because P(x) \wedge Q(x)$

$\therefore x^2=x$

$x^2-x=0$

$x(x-1)=0$

$x_1=0 \quad x_2=1$

Also $\because x \le 0$

$\therefore x=0$

$\therefore x$ is even $\quad \therefore S(x) \quad \therefore \forall x \in \mathbb{Z}, (P(x) \wedge Q(x)) \to S(x)$

Thus the statement is true. $\square$

(g) False

$\because R(x)$

$\therefore x^2 = x+1$

$x^2 - x - 1 = 0$

$a=1 \quad b=-1 \quad c=-1 \quad \Delta = b^2-4ac = 1+4 = 5 > 0$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$= \frac{1 \pm \sqrt{5}}{2} \notin \mathbb{Z}$

Thus the statement is false.

(h) True

Proof: Take $x=-2 \in \mathbb{Z}$,

$\therefore x$ is even

$\therefore S(x)$

$\because x=-2 \le 0$

$\therefore Q(x)$

$\therefore S(x) \wedge Q(x)$

$\therefore \exists x \in \mathbb{Z}$ such that $S(x) \wedge Q(x)$ $\square$

(i) False

Take $x=1 \in \mathbb{Z}, y \in \mathbb{Z}$

Then $x<y \Rightarrow y \in \mathbb{Z}$

But $y \ge 2$

$\therefore y^2 > x^2$

$\therefore \neg T(x,y)$

Thus the statement is false.

(j) True

Proof: $\because \sim P(x)$

$\therefore x^2 \ne x$

$x^2 - x \ne 0$

$x(x-1) \neq 0$

$x \neq 0$ or 1 Also $\because x \in \mathbb{Z}$ $\therefore x \leq -1$ or $x \geq 2$

Suppose $x < y$, $x, y \in \mathbb{Z}$

$\therefore y \geq x+1$

$x+1 < x^2$

$x^2 - x - 1 > 0$

$a = 1$ $b = -1$ $c = -1$

$\Delta = b^2 - 4ac = 1 + 4 = 5 > 0$

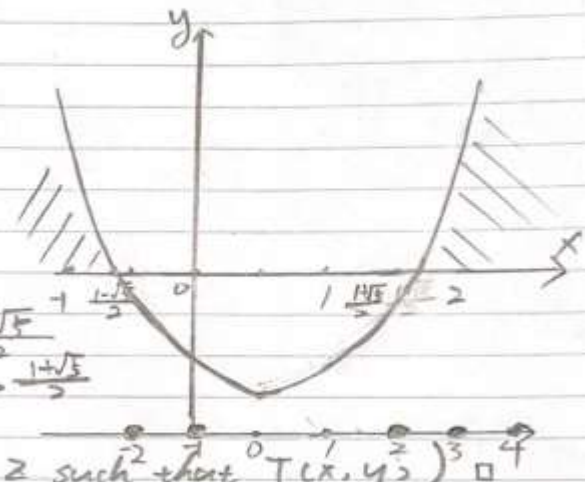
$x = \frac{-b \pm \sqrt{\Delta}}{2a}$

$= \frac{1 \pm \sqrt{5}}{2}$

$\therefore x < \frac{1-\sqrt{5}}{2}$ or $x > \frac{1+\sqrt{5}}{2}$

Also $\because x \leq -1$ or $x \geq 2 > \frac{1+\sqrt{5}}{2}$

$\therefore T(x, y)$



$\therefore \forall x \in \mathbb{Z}, (\sim p(x)) \rightarrow \exists y \in \mathbb{Z}$ such that $T(x, y))$ $\square$